



Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member (s): Dr.R.Venkatesh, AP (SG)/CSE

Date: 23.09.2021

Innovative Practice Description

- **Unit / Topic:** Unit II / Advanced Encryption Standard
- **Date of Implementation:** 15.09.2022
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Think-Pair-Share
- **Justification:**
 - Because Advanced Encryption Standard Implementation Process and detailed structure is such an interesting and important topic, the students are eager to discuss and share their ideas. This practice assists students in gaining a thorough understanding of this topic and real-time application projects. Students think about a topic on their own and express their thoughts with their classmates, which helps them enhance their creative thinking and oral communication abilities.

Time Allotted for the Activity:

- **Procedure of the Implementation:**
 - Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
 - T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-5 Minutes].
 - P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 2-5 Minutes].
 - S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 05-10 minutes].

Details of the Implementation

- First of all, teacher taught the topics and provide the materials to think and discuss.
- Teacher provide a different application based questions about the topic and ask them to form a group and gave a 5 minutes' time to think. The Students come up with their own individual answer to the question.



- Finally, the students shared their ideas and answers based on their application based questions.
- The Figure 1 shows the Think Pair Share activity done by the students.
- Finally, teacher and students discussed the answers of all questions at the end of the activity.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO8	PO10	PO11	PO12
CO1	3	3	3	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

- Images / Screenshot of the practice:



Figure 1 – Students Think and Share their Thoughts and Topic Presentation

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ❖ Most of the students actively participated and enjoyed. Students were given a solution with various formats.

- ❖ Bright students enjoyed with peer learning.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students are actively engaged.
- Students learn from each other.
- Makes class interactive.



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- Builds a friendly, yet academic atmosphere.
- Includes all the students in the teaching-learning process.

❖ *Challenges faced in implementation:*

- Students were not aware of TPS even though made an announcement and posted the content of TPS in the course web site.
- Slow learners were not interested to participate in TPS.
- Noise level of the class room goes up and a transition was needed to get back on the topic.
- It was hard to keep every student on task.
- It took a longer time than other strategies.

References:

1. adlit.org, “Adolescent Literacy”, 2019. [Online]. Available: <http://www.adlit.org/strategies/23277/> [Accessed: 23.01.2020]
2. Reading Books - <https://www.readingrockets.org/strategies/think-pair-share>
3. prodigygame.com, “active-learning-strategies”, 2019. [Online]. Available: <https://www.prodigygame.com/blog/active-learning-strategies-examples/>
4. Mr.R.Venkatesh., “Introduction to Yoga and meditation for professional excellence and stress management.” in course GE6075 Professional Ethics in engineering <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>

Signature of Faculty Member

HOD-CSE